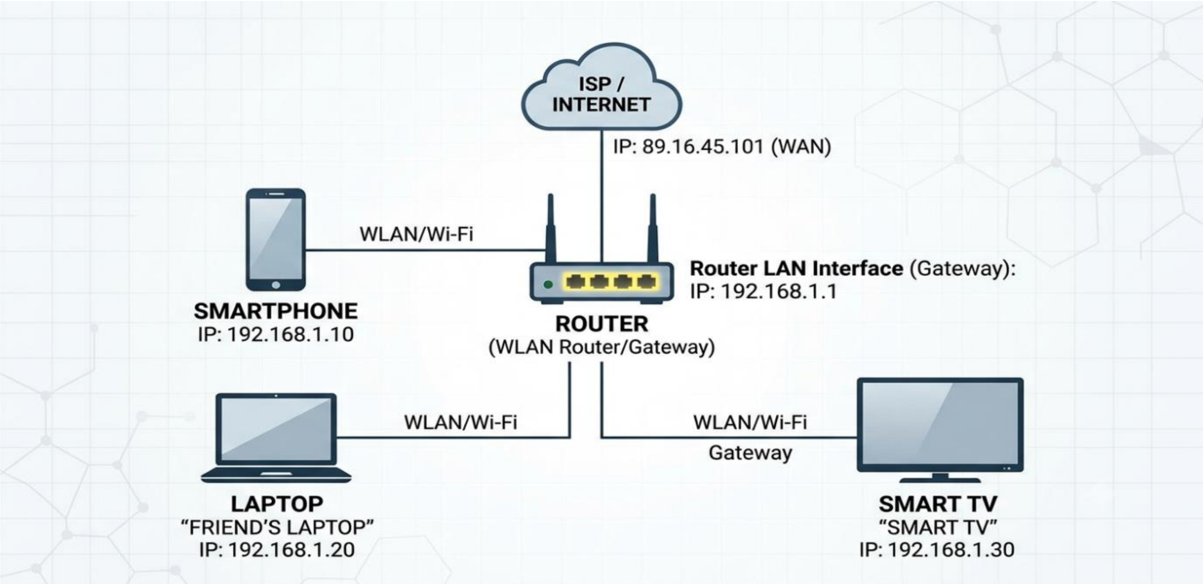
1. **Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.**



2. **Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.**

- **Manual IP routing**

Laptop

IP: 192.168.1.10/24

Gateway: 192.168.1.1

Router R1

LAN: 192.168.1.1

WAN: 10.0.0.1

Router R2

WAN: 10.0.0.2

LAN: 192.168.2.1

Friend's Computer

IP: 192.168.2.20/24

Gateway: 192.168.2.1

- **IP addresses of devices**

Device	**IP address**	**Subnet Mask**
Laptop	192.168.1.10	255.255.255.0
Router R1(LAN)	192.168.1.1	255.255.255.0
Router R1(WAN)	10.0.0.1	255.255.255.252
Router R2(WAN)	10.0.0.2	255.255.255.252
Router R2(LAN)	192.168.2.1	255.255.255.0
Friend's computer	192.168.2.20	255.255.255.0

- **Steps of Packet Travel**
- **Step-1 Laptop Creates the Packet**
- Source IP: 192.168.1.10 and destination IP: 192.168.2.20
- The laptop compares its own subnet mask with the destination subnet.
- Since they are different, the laptop knows the destination is on another network.
- It forwards the packet to its default gateway which is 192.168.1.1
- **Step-2 Laptop Checks it Routing Table**

Destination network	**Netmask**	**Gateway**	**Interface**
192.168.1.0	255.255.255.0	Direct	Ethernet
0.0.0.0	0.0.0.0	192.168.1.1	Ethernet

Here, no direct route exists for 192.168.2.20, so default route would be 192.168.1.1

- **Step-3 Router R1 Receives the Packet**
- Router R1 examines the destination IP address

Routing R1 Routing Table

Destination Network	**Netmask**	**Next Hop**	**Interface**
192.168.1.0	255.255.255.0	Direct	LAN
10.0.0.0	255.255.255.252	Direct	WAN
192.168.2.0	255.255.255.0	10.0.0.2	WAN

- Here, destination belongs to network 192.168.2.0/24 and It forward the packet to the next hop which is 10.0.0.2 of Router R2
- **Step-4 Router R2 Receives the Packet**
- In this step, router R2 checks its routing table.

Router R2 Routing Table

Destination Network	**Netmask**	**Next Hop**	**Interface**
192.168.2.0	255.255.255.0	Direct	LAN
10.0.0.0	255.255.255.252	Direct	WAN
192.168.1.0	255.255.255.0	10.0.0.1	WAN

- Here, the destination IP address 192.168.2.20 is directly connected with LAN.
- Router R2 uses ARP to find the MAC address of 192.168.2.20.
- And the packet is forwarding directly to the friend’s computer.
- **Step-5 Friend’s Computer Receives the Packet**
- The friend’s computer receives the Ethernet frame.
- It checks destination IP 192.168.2.20 and matching it with its own IP address.
- Lastly, packet is accepted and passed it to the web browser.

3. **Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.**

- Here is routing table for fictional Zomato-style delivery network:

Network Design

Branch	**City**	**Network Address**	**Subnet Mask**	**Gateway Router**
Branch 1	Ahmedabad	192.168.10.0	255.255.255.0	R1
Branch 2	Mumbai	192.168.20.0	255.255.255.0	R2
Branch 3	Bengaluru	192.168.30.0	255.255.255.0	R3

Routing Table (Headquarter Router)

Destination Network	**Subnet Mask**	**Next Hop**	**Interface**
192.168.10.0	255.255.255.0	10.0.0.2	GigabitEthernet0/0
192.168.20.0	255.255.255.0	10.0.1.2	GigabitEthernet0/1
192.168.30.0	255.255.255.0	10.0.2.2	GigabitEthernet0/2
10.0.0.0	255.255.255.252	Directly Connected	GigabitEthernet0/0
10.0.1.0	255.255.255.252	Directly Connected	GigabitEthernet0/1
10.0.2.0	255.255.255.252	Directly Connected	GigabitEthernet0/2

Example

- If a delivery management server in the Ahmedabad branch wants to communicate with a billing server in the Mumbai branch.

a) The packet reaches the HQ router.
b) The HQ router checks it routing table.
c) It finds the route for 192.168.20.0
d) The packet is forwarded to the next hop 10.0.1.2 through GigabitEthernet0/1
e) Router R2 delivers the packet to the Mumbai branch network.

4. **Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).**

- **Static Routing Advantages**
- Simple to configure for small networks.
- Low CPU and memory usage because routers do not exchange routing updates.
- More secure since routes are manually configured and cannot be changed automatically.
- **Static Routing Disadvantages**
- Requires manual updates whenever the network changes.
- Does not automatically recover from link failures.
- Not suitable for large networks because managing many routes is time-consuming.
- **Real-World Example**
- A small office with one internet connection can use static routing because the network rarely changes.
- **Dynamic Routing Advantages**
- Automatically learns and updates routes when the network changes.
- Scales well for medium and large networks.
- Provides redundancy by finding alternate path if a link fails.
- **Dynamic Routing Disadvantages**
- More complex to configure and troubleshoot.
- Uses more CPU, memory and bandwidth due to routing updates.
- May take time to converge after network changes.
- **Real-World Example**
- A food delivery app like Zomato or Uber Eats benefits from dynamic routing because if one network path fails, traffic is automatically redirected through another path, keeping the service available.

5. **Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.**

- Below is a simple Cisco configuration for setting up OSPF routing between two routers and each line has its explanation.

Router 1 – R1 Configuration

```
Enable
Configure terminal
Hostname R1
Interface GigabitEthernet0/0
 Ip address 192.168.1.1 255.255.255.0
 No shutdown
Interface GigabitEthernet0/1
 Ip address 10.0.0.1 255.255.255.252
 No shutdown
Router ospf 1
 Router-id 1.1.1.1
 Network 192.168.1.0 0.0.0.255 area 0
 Network 10.0.0.0 0.0.0.3 area 0
End
Write memory
```

Configuration Line	Explanation
enable	Enters privileged EXEC mode so you can make configuration changes.
Configuration terminal	Opens global configuration mode.
Hostname R1	Sets the router's name to R1.
Interface GigabitEthernet0/0	Selects interface G0/0 for configuration.
Ip address 192.168.1.1 255.255.255.0	Assigns the LAN IP address to G0/0.
No shutdown	Enables the interface. By default, interfaces are administratively down.
Interface GigabitEthernet0/1	Selects the interface connecting to router 2.
Ip address 10.0.0.1 255.255.255.252	Assigns an IP address to the point-to-point link.
No shutdown	Activates the interface.
Router ospf 1	Starts ospf process number 1. The process ID is locally significant.
Router-id 1.1.1.1	Gives R1 a unique ospf router ID.
Network 192.168.0.0 0.0.0.255 area 0	Advertises the LAN network into ospf area 0.
Network 10.0.0.0 0.0.0.3 area 0	Advertises the link between R1and R2 into ospf area 0.
end	Exits configuration mode.
Write memory	Saves the configuration so it remains after reboot.

Router 2 – R2 Configuration

Enable

Configure terminal

Hostname R2

Interface GigabitEthernet0/0

Ip address 192.168.2.1 255.255.255.0

No shutdown

Interface GigabitEtheret0/1

Ip address 10.0.0.2 255.255.255.252

No shutdown

Router ospf 1

Router-id 2.2.2.2

Network 192.168.2.0 0.0.0.255 area 0

Network 10.0.0.0 0.0.0.3 area 0

End

Write memory

Configuration line	Explanation
enable	Enters privileged EXEC mode.
configure terminal	Opens global configuration mode.
hostname R2	Sets the router name to R2.
interface GigabitEthernet0/0	Selects the LAN interface.
ip address 192.168.2.1 255.255.255.0	Assigns the LAN IP address.
no shutdown	Enables the interface.
interface GigabitEthernet0/1	Selects the interface connected to R1.
ip address 10.0.0.2 255.255.255.252	Assigns the IP address for the router-to-router link.
no shutdown	Activates the interface.
router ospf 1	Starts OSPF process 1.
router-id 2.2.2.2	Assigns a unique Router ID for R2.
network 192.168.2.0 0.0.0.255 area 0	Advertises R2's LAN network into OSPF.
network 10.0.0.0 0.0.0.3 area 0	Advertises the shared link with R1.
end	Exits configuration mode.
write memory	Saves the running configuration.